

Milestone 1

Team Formation and Planning

Execution Document — Tasks, Timeline and Ownership

AI & Data Science — Microsoft Machine Learning Engineer Track

Digital Egypt Pioneers Initiative (DEPI) — Graduation Project

What this document is. The execution plan for the first DEPI milestone: every task, who owns it, who works on it with them, how many hours it should take, what it depends on, and what it produces. It is written to be worked from day by day, not read once. It is the first of seven such documents, one per milestone.

The team. **Aliaa Gomaa** (M1, Data Engineering & Experiment Validity) · **Mohammed Sherif** (M2, Uplift Modeling Core) · **Mahmoud Nadeem** (M3, Advanced Models & Experiment Tracking, and Team Leader) · **Ali Alaa** (M4, Evaluation, Statistics & Decision Layer) · **Mazen Mahmoud** (M5, MLOps, Backend & Product). The role codes M1–M5 appear in the task tables for compactness; Section 4 lists what each one owns and Section 11 gives the hour load per person.

Every task in this document has two people on it. One is the owner, who does the work and is accountable for it; the second reviews it, tests it, or works alongside them. This is deliberate and it costs roughly 25% more hours than assigning one person per task. It buys three things: nothing in the project ends up understood by only one person, every artefact is seen by a second pair of eyes before anyone builds on it, and each member touches parts of the system outside their own role — which is the point of doing the project at all.

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1 Milestone Identity

Milestone	1 of 7 — Team Formation and Planning
DEPI step	“Project Planning” — the first stage of the official project plan
Programme weeks	Week 5 and Week 6
Hard deadline inside it	The Project Proposal is submitted to the instructor by the end of Week 5. Week 6 is the technical foundation and carries no external deadline.
Owner	Team Leader
Total effort	Approximately 88 person-hours across five members and two weeks, which is about 8–9 hours per member per week
Blocks	Milestone 2 (Instructor Review, Week 7) cannot happen without the proposal produced here

Why this milestone spans two weeks rather than one. The DEPI plan marks team formation at Week 5 and the instructor review at Week 7, leaving Week 6 unlabelled. Treating Week 6 as part of this milestone rather than as a gap is what makes the Week 7 review substantive: the team walks in with a working environment, verified data and a first chart, instead of walking in with a document. That difference changes the tone of the review and the quality of the feedback received.

2 What This Milestone Actually Decides

It looks administrative. Two of the decisions taken here are not, and they are expensive to reverse later.

1. **Whether the datasets are what the documentation says they are.** The team verifies the published figures against the actual files in this milestone. If they disagree, every number in the proposal and the specification changes, and the team needs to know that in Week 5 rather than in Week 14.
2. **Whether the whole team understands the core question.** If only one or two members can explain why classification accuracy does not apply to this problem, the project has a single point of failure in its own reasoning — and the final defence asks every member, not one.

3 Entry and Exit Criteria

3.1 Entry criteria

- Project direction agreed: uplift modelling, standalone (no mega-project merge).
- The master specification exists and is available to all five members.
- The five members are identified.

3.2 Exit criteria — the definition of done

1. Project Proposal submitted to the instructor, and the submission recorded with a date.
2. Every member can state, unprompted, the project's core question and why accuracy is not its metric.
3. Role numbers mapped to named people, written down, and circulated.
4. Repository exists, is structured, and continuous integration runs green on a trivial commit.
5. The shared environment installs and runs on all five machines — verified by all five, not assumed.
6. Both datasets downloaded, checksummed, and their published figures verified against the files, with any discrepancy written down.
7. KPI definitions, risk register and project plan exist as documents in the repository.
8. MLflow logs a run locally.
9. One chart exists — any chart, from the Hillstrom data.

Point 9 is not decoration. Walking into the Week 7 instructor review with a plot made from the project's own data changes the meeting from a plan review into a progress review.

4 Roles and Ownership

Number	Role	Owns across the project
Aliaa Gomaa (M1)	Data Engineering & Experiment Validity	Ingestion, schema, randomization integrity, leakage audit, the split, preprocessing, the ERD
Mohammed Sherif (M2)	Uplift Modeling Core	Baselines, S/T/X-learners, the class-transformation demonstration
Mahmoud Nadeem (M3)	Advanced Models & Experiment Tracking	Uplift forest, neural estimator, Optuna, MLflow tracking and registry
Ali Alaa (M4)	Evaluation, Statistics & Decision Layer	Metric harness, bootstrap intervals, negative control, decision layer, abstention, simulator
Mazen Mahmoud (M5)	MLOps, Backend & Product	API, Docker, Azure, database, dashboard, monitoring, CI/CD, replay scenario plumbing
Team Leader <i>Mahmoud</i>	Held <i>in addition</i> to the M3 role	Proposal, plan, submissions, repository ownership, speaking for the team

The critical path is Aliaa (M1) then Ali (M4). Nothing anyone else produces is trustworthy until the data and the split are valid, and no model can be compared with another until the evaluation harness with confidence intervals exists. Those two roles carry the credibility of everything else in the project.

Two tasks below are marked $\rightleftharpoons$: they are written to be handed to another member without losing context, if anyone's week runs over.

5 How Two People Work on One Task

Every task has an **Owner** and a **Partner**. They are not the same relationship in every case, and the table says which is meant:

Mode	What the partner actually does
Review	The owner produces it; the partner reads it critically against a stated checklist and signs off in the pull request. Used for documents and for decisions.
Pair	Both sit on it together for part of the work — typically the hard or unfamiliar part — then the owner finishes alone. Used for setup work where a second person needs to be able to do it later.
Verify	The partner independently repeats the check on their own machine and confirms the result matches. Used where being wrong is expensive: dataset figures, environment installs.
Consume	The partner is the next person in the chain, and joins at the end to accept the handover and confirm the output is in the shape they need.

The partner is chosen from a *different* role wherever possible. Pairing Aliaa with Mohammed on the dataset work means the modelling member knows how the data was built; pairing Mazen with Mahmoud on the repository means the tracking work fits the structure it will live in. Spreading knowledge sideways like this is the reason the pairing exists — reviewing alone would be cheaper.

6 Work Breakdown — Week 5 (Proposal Sprint)

ID	Task	Owner	Partner	Hours	Needs	Output
T1	Read the specification: Parts I–III in full, IV–VI skimmed	All five	—	2.5 each (12.5)	—	Shared understanding; questions list
T2	Kickoff: working agreement, meeting cadence, communication channel, definition of done	Mahmoud (TL)	All (pair)	1.5 each (7.5)	T1	Working-agreement note in the repository
T3	Map role numbers to people; record and circulate	Mahmoud (TL)	All (review)	1.0	T2	Role table

ID	Task	Owner	Partner	Hours	Needs	Output
T4	Repository skeleton: folder structure of the specification, README stub, <code>.gitignore</code> , branching strategy, coding standards	Mazen	Mahmoud (pair)	4.0 + 1.5	T3	Repository on the organisation account
T5	Request the project repository from the company; invite all members	Mahmoud (TL)	—	0.5	T3	Repository access for five
T6	Download both datasets; record checksums; write the loader script; draft the dataset cards	Aliaa	Mohammed (con-sume)	4.0 + 1.5	T4	<code>loaders.py</code> , checksums, dataset cards
T7	KPI confirmation: work through the eleven KPIs in the specification, confirm each success condition, and fix the threshold values they depend on in <code>config</code>	Ali	Mohammed (review)	3.0 + 1.0	T1	<code>docs/kpis.md</code>
T8 $\rightleftharpoons$	Risk register v1: risk, severity, mitigation, owner, trigger	Mazen	Ali (review)	2.0 + 1.0	T1	<code>docs/risks.md</code>
T9 $\rightleftharpoons$	Project plan and Gantt chart across programme weeks 5–21	Mahmoud (TL)	Mohammed (review)	2.0 + 1.0	T3, T7	<code>docs/plan.md</code> + Gantt image
T10	Write the Project Proposal on the official DEPI template	Mahmoud (TL)	Ali (pair)	3.5 + 2.0	T3, T7, T9	<code>Proposal.docx</code> draft
T11	Internal review: all five read the proposal against the DEPI template checklist and comment	All five	—	1.0 each (5.0)	T10	Comment list
T12	Revise and submit; record the submission date and channel	Mahmoud (TL)	—	1.5	T11	Submitted proposal

Week 5 total: approximately 50 person-hours.

7 Work Breakdown — Week 6 (Foundation Sprint)

ID	Task	Owner	Partner	Hours	Needs	Output
T13	Environment: <code>pyproject.toml</code> , pinned dependencies, Python version, install instructions	Mahmoud	Mazen (pair)	3.0 + 1.0	T4	Reproducible environment
T14	Environment verified independently on every machine	All five	—	0.5 each (2.5)	T13	Five confirmations recorded
T15	CI stub: lint, format check, type check, one trivial test, running on pull requests	Mazen	Aliaa (review)	3.0 + 1.0	T4	Green CI badge
T16	Verify the published dataset figures against the actual files: row counts, column lists, treatment ratio, visit and conversion rates	Aliaa	Ali (verify)	2.5 + 1.0	T6, T13	Verification report; discrepancies flagged
T17	Unit-of-observation check: what does one row represent, and do identifiers repeat?	Aliaa	Mohammed (review)	2.0 + 1.0	T16	Finding recorded, with its consequence for splitting
T18	Tooling decision: <code>scikit-uplift</code> versus <code>causalml</code> — what each provides, what is used for which purpose, versions pinned	Mohammed	Mahmoud (review)	2.5 + 1.5	T13	Decision note in <code>docs/</code>
T19	MLflow running locally, logging a dummy run with parameters, metrics and an artefact	Mahmoud	Mazen (pair)	2.5 + 1.0	T13	Working tracking server; screenshot
T20	First exploratory analysis of Hillstrom: arm sizes, outcome rates per arm, and one chart	Ali	Aliaa (con- sume)	2.5 + 1.0	T16	<code>01_explore</code> notebook + one figure
T21	Causal vocabulary note for the team: potential outcomes, ATE, CATE, ITT, SUTVA, overlap, Qini — one paragraph each	Mohammed	Ali (review)	2.0 + 0.5	T1	<code>docs/glossary.md</code>

ID	Task	Owner	Partner	Hours	Needs	Output
T22	Milestone retrospective: what took longer than planned, and what Milestone 2 needs	Mahmoud All (TL)		0.5 each (2.5)	—	Retro note

Week 6 total: approximately 38 person-hours.

T20 is placed on Ali on purpose. The evaluation role is the one that spends the most of the project working with numbers rather than with data, and putting the first exploratory look in that person's hands early means the metric harness they build later is informed by what the data actually looks like. Aliaa joins at the end as the person who produced the data, to confirm the reading is right.

8 Timeline — Day by Day

Fourteen working days across the two weeks. Vertical position is time; horizontal bars run in parallel. Tracks are independent unless an arrow says otherwise.

WEEK 5 -- PROPOSAL SPRINT						deadline: proposal submitted
DAY	ALL	DATA (M1,M2)	BUILD (M5,M3)	DOCS (TL,M4)	METHOD (M4,M2)	
1	T2 kickoff					
2	T1 reading	T1	T1	T1	T1	
3			T4 repo ####	T3 roles		
				T5 repo request		
4		T6 download	T4 repo ####		T7 KPIs ####	
		#####				
5		T6 #####	----> ready	T9 plan+Gantt	T7 ####	
6			T8 risks ####	T10 proposal ####		
7	T11 review			T12 revise+SUBMIT		
WEEK 6 -- FOUNDATION SPRINT						no external deadline
DAY	ALL	DATA (M1,M2)	BUILD (M5,M3)	METHOD (M2,M3)	ANALYSIS (M4,M1)	
8			T13 env #####			
9	T14 verify	T15 CI (M1	T13 env ---->			
	env	reviews)	T15 CI #####			
10		T16 verify		T18 tooling ####		
		figures				
11		T17 unit-of-		T18 ####		
		observation		T19 MLflow #####		
12				T21 glossary	T20 explore +	
					first chart	
13		BUFFER -- catch-up on anything that slipped				
14	T22 retro	Milestone 2 readiness check				

Days 1–2 are deliberately spent reading. Two days of five people reading looks like two days lost. It is the cheapest insurance in the project: the failure mode it prevents — three members who never internalised why accuracy does not apply, discovered during the final defence — cannot be fixed later at any price.

9 Handover Chain — Who Waits on Whom

```

LEADER  T3 role mapping
|
+--> M5  T4 repository skeleton
|
|      +--> M3  T13 environment ---> ALL  T14 verify on every machine
|      |
|      |      +--> M3  T19 MLflow running
|      |      +--> M2  T18 tooling decision
|      |
|      +--> M5  T15 CI pipeline (M1 reviews)
|
+--> M1  T6 download datasets
|
|      +--> M1  T16 verify published figures (M4 verifies independently)
|      |
|      |      +--> M1  T17 unit-of-observation check
|      |      +--> M4  T20 first exploration + first chart
|
+--> M4  T7 KPI definitions
|
|      +--> LEADER  T9 project plan and Gantt
|      |
|      |      +--> LEADER + M4  T10 proposal
|      |      |
|      |      |      +--> ALL  T11 review
|      |      |      |
|      |      |      |      +--> LEADER T12 SUBMIT

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9.1 The two chains that must not stall

Chain	Why
T3 → T4 → T13 → T14	Everything technical in Week 6 waits on the environment. If the repository is not up by Day 5, the whole second week compresses.
T3 → T7 → T9 → T10 → T12	The proposal chain, and it ends at an external deadline. If the KPI definitions are not done by Day 5, the proposal is written without them and has to be revised after submission.

10 What Runs in Parallel

Track	Who	Runs independently of
Data	Aliaa owns, Mohammed partners	The build track and the docs track entirely. Only the environment (T13) creates a dependency, and only from Week 6.
Build	Mazen owns, Mahmoud partners	The data track entirely.
Docs and planning	Mahmoud (TL) owns, Ali partners	Both technical tracks; needs only the role mapping.
Method	Mohammed with Mahmoud; Ali with Mohammed	Needs the environment; otherwise independent.
Analysis	Ali owns, Aliaa partners	Needs verified data; last in the Week 6 sequence.

Three tracks run simultaneously through most of Week 5 and four through Week 6. Nobody in this plan is ever waiting with nothing to do, and no task has more than one upstream dependency — which is what makes the schedule survive one person being unavailable for two days.

11 Effort Summary

Role	Week 5	Week 6	Total	Heaviest items
Aliaa (M1)	9.0 h	7.5 h	16.5 h	Dataset download (T6), figure verification (T16)
Mohammed (M2)	8.5 h	6.5 h	15.0 h	Tooling decision (T18), plan review (T9)
Mahmoud (M3)	6.5 h	8.0 h	14.5 h	Environment (T13), MLflow (T19)
Ali (M4)	11.0 h	5.0 h	16.0 h	KPI definitions (T7), proposal support (T10)
Mazen (M5)	11.0 h	6.0 h	17.0 h	Repository (T4), CI (T15)
Team Leader (<i>overhead, on top of a role</i>)	8.5 h	0.5 h	9.0 h	Proposal (T10), plan (T9), submission (T12)
Team total	≈50 h	≈38 h	≈88 h	

11.1 If the team has fewer hours than this

The plan assumes 8–9 hours per member per week. If that is not realistic, cut in this order and record the decision:

1. T21 glossary — valuable, not blocking.
2. T20 exploration and first chart — costs the Week 7 review its strongest moment, so cut it last among the optional items.
3. T18 tooling decision — can be deferred to Week 7 if the environment is up.

Never cut: T1 (reading), T14 (environment verified by everyone), T16 (figure verification), or anything on the proposal chain.

12 Deliverables

12.1 Required by DEPI for this stage

Deliverable	Task	Form
Project Proposal	T10, T12	.docx on the official template: objective, description, tools and design, technologies, members and responsibilities, weekly tasks and deliverables
Project Plan	T9	Timeline with milestones, deliverables and a Gantt chart
Task Assignment & Roles	T3	Named table, submitted with the proposal
Risk Assessment & Mitigation	T8	Risk register with severity, mitigation and an owner per risk
KPIs	T7	The eleven KPIs confirmed, with their thresholds fixed in config before any result is produced

12.2 Technical artefacts produced alongside

Artefact	Task	Acceptance test
Repository with structure	T4	The folder layout matches the specification; a new member can find where anything belongs without asking
Reproducible environment	T13, T14	Installs cleanly on all five machines from the written instructions
Continuous integration	T15	Runs on a pull request and goes green
Dataset loaders and cards	T6	<code>load_hillstrom()</code> and <code>load_criteo()</code> return data from a clean checkout
Figure verification report	T16	Every published statistic either confirmed or flagged, in writing
Unit-of-observation finding	T17	A stated answer to “what is one row?” with its consequence for splitting
Tooling decision note	T18	Which library is used for what, with versions pinned
MLflow tracking	T19	One run visible in the UI with parameters, a metric and an artefact
First exploration and chart	T20	Arm sizes and outcome rates per arm, and one figure
Glossary	T21	Every term used in the specification has a one-paragraph entry

13 Risks Specific to This Milestone

Risk	Severity	Mitigation
Only one or two members genuinely read the specification	High	T1 is scheduled as work, not homework, and T2 ends with each member stating the core question in their own words. If someone cannot, the reading is not finished.
Published dataset figures do not match the files	Medium	T16 exists to find this in Week 6. If it happens, the proposal and specification are corrected immediately and the discrepancy is documented — which is a stronger position than never having checked.
The team leader is overloaded in Week 5	High	Week 5 carries the proposal on top of a member role. The two tasks marked $\rightleftharpoons$ (T8, T9) are written to be handed to another member — hand them off early rather than at the deadline.
Repository access from the company is slow	Medium	T5 is requested on Day 3, not Day 7. Work proceeds on a temporary repository and history is pushed across when access arrives.
Environment works for some members and not others	Medium	T14 requires an independent confirmation from each of the five. “It works on mine” is not a confirmation.
The proposal is written the night before the deadline	Medium	The chain $T7 \rightarrow T9 \rightarrow T10$ is scheduled to finish on Day 6, leaving Day 7 for review and revision rather than for writing.

14 What Each Member Should Be Able to Say at the End

Not a formality — the final DEPI evaluation asks each member what they did, and the answer is built here, not remembered later.

Role	The sentence
Aliaa Gomaa	“I built the data loaders and then checked the published figures against the actual files — including what one row of the dataset really represents, which decides how we are allowed to split it.”
Mohammed Sherif	“I chose the causal-inference libraries and wrote the vocabulary note the rest of the team works from.”
Mahmoud Nadeem	“I built the environment and the experiment tracking, so that every result we report later can be traced back to the configuration that produced it.”
Ali Alaa	“I defined what success means for this project as measurable numbers, and produced the first look at the data.”
Mazen Mahmoud	“I built the repository, the standards and the CI pipeline that every later change goes through.”
Mahmoud, as leader	“I mapped the roles against the critical path, wrote the proposal, and ran the milestone.”

Every one of those sentences describes something the person *did*, not something they *were assigned*. If at the end of Week 6 a member cannot say one of them truthfully, the milestone is not finished for that member — regardless of what the checklist says.

15 Sign-off Sheet

Checked in the retrospective (T22). Every line is either yes or a written reason.

Done	Item
☐	Proposal submitted; date and channel recorded
☐	All five members can state the core question and why accuracy is not the metric
☐	Role numbers mapped to people, written down, circulated
☐	Repository exists with the agreed structure; CI green
☐	Environment verified independently on all five machines
☐	Both datasets downloaded and checksummed; loaders work from a clean checkout
☐	Published figures verified against the files; discrepancies written down
☐	Unit-of-observation question answered in writing
☐	KPI definitions, risk register and project plan committed to the repository
☐	MLflow logs a run locally
☐	One chart exists from the Hillstrom data
☐	Retrospective written: what slipped, and what Milestone 2 needs

16 Handover to Milestone 2

Milestone 2 is the **Instructor Review in Week 7**, and it is a one-week milestone with a single external event in it. What it inherits from here:

- The submitted proposal, which is what is being reviewed.
- A working environment and repository, so that the review can show progress rather than intention.
- The first chart from T20 — the single artefact that changes the character of the meeting.
- The figure-verification report, which is the evidence behind every number quoted in the proposal.

Carry these four questions into Milestone 2 and come out with written answers. They are listed here because they must be prepared during Week 6, not improvised in the meeting.

1. Are computer vision and NLP *required* components of the evaluation, or desirable ones? This project has neither, deliberately, and Week 7 is the last point at which that answer can change the project cheaply.
2. Is a public benchmark dataset acceptable as the primary data source, or is locally collected data expected?
3. Is the project graded per team, and does any part of the grade depend on another team?

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4. What weight do documentation, testing and deployment carry relative to model results?
-

Milestone 1 of 7. Hour estimates are planning figures for a part-time student team at roughly 8–9 hours per member per week; record the actual hours in the retrospective and adjust the remaining milestone documents against the real numbers rather than these ones.